

Software Requirement Specification

Functional Requirements

1. The system shall have a GUI window that opens when the user runs main.py.
2. The GUI shall have a text area where the user can type or paste BasicML code directly.
3. The GUI shall show program output in a console area on the left side of the window
4. The GUI shall have a press enter or click submit button to send user input (BasicML program) into the running program.
5. The system shall have a fixed size memory array.
6. The system shall support all 12 BasicML opcodes: READ WRITE LOAD STORE ADD SUBTRACT DIVIDE MULTIPLY BRANCH BRANCHNEG BRANCHZERO and HALT
7. When the BasicML program uses the READ opcode the GUI shall have an input box so the user can enter a number.
8. The system shall validate the format of the loaded program before execution.
9. The system shall display an error message when a program contains invalid instructions.
10. Errors like division by zero, invalid opcodes, or invalid memory access shall be detected by the system.
11. Error messages like division by zero, invalid opcodes, or invalid memory access shall be shown in red in the console.
12. The system shall give an error message if no program is loaded.
13. The system shall display the current value of the accumulator.
14. The GUI shall have a reset button that clears the console and resets the accumulator back to 0.
15. The system shall show a message in the console when the program finishes running or halts.

Non-Functional Requirements

1. The GUI shall be intuitive and easy to use without instructions

2. All controls shall be clearly labeled
3. The shall will handle invalid input without crashing
4. The codebase shall be modular and easy to update
5. The system shall run on any platform supporting Python

Error Handling

The system shall handle the following errors:

1. Invalid instruction format
2. Invalid opcode
3. Memory access out of bounds
4. Division by zero
5. Invalid color file input

Summary

The BasicML Application provides a complete environment for writing, editing, and executing BasicML programs. It supports multiple instruction formats, includes strong validation, and is user friendly.